perplexity

What BCIs can actually do

As of 2026, BCIs can reliably turn neural activity into digital commands to **restore lost function**, **control devices**, and **monitor cognitive states**, with most real-world use concentrated in healthcare and neurorehabilitation. [1] [2] [3]

Medical and clinical capabilities (proven today)

These are the areas where BCIs are already working in real patients or tightly monitored trials:

- **Communication for paralysis and ALS**
  - Paralyzed users can **type text, control cursors, and select icons** using only neural signals. [3] [4]
  - Systems can decode **intended speech, handwriting, or text selection** to enable communication for people who cannot speak or move. [4] [5]
- **Restoring and controlling movement**
  - BCIs drive **robotic arms, exoskeletons, and functional electrical stimulation (FES)** systems so paralyzed users can grasp, reach, and perform basic tasks. [1] [3]
  - Amputees can control **advanced prosthetic limbs** with multiple degrees of freedom and receive **sensory feedback** (e.g., feeling touch or pressure). [2] [3]
- **Neurorehabilitation after stroke and spinal cord injury**
  - BCI-based therapy helps patients **retrain movement** by pairing brain signals with robotic or electrical assistance, improving motor recovery in some stroke and spinal cord injury cohorts. [2] [3]
- **Neuromodulation for mental health and pain**
  - Closed-loop systems that read brain activity and deliver stimulation are being used or tested to **alleviate depression, PTSD, chronic pain, and anxiety** in early clinical programs. [6] [3]
- **Seizure detection and control**
  - Devices like NeuroPace's RNS (already FDA-approved) detect abnormal brain activity and deliver **on-demand stimulation** to reduce seizures, showing the feasibility of bidirectional BCIs. [7] [6]

Assistive and environmental control

BCIs are also used to control external systems to improve independence:

- **Smart home and environmental control**
  - Users can operate **lights, thermostats, TVs, and other smart-home devices** via neural commands, especially helpful for people with severe motor limitations. [3] [1]
- **Mobility devices**
  - Experimental and early commercial systems allow users to control **wheelchairs, mobility platforms, and navigation aids** using brain signals, often combined with eye-tracking or other sensors. [1] [3]
- **Computer and device control**
  - Invasive BCIs (e.g., Neuralink, Synchron, Blackrock) enable patients to **control iPads, PCs, and other digital interfaces** for browsing, messaging, and basic productivity from home. [4] [7]

Cognitive and performance monitoring (non-invasive BCIs)

Non-invasive EEG/fNIRS wearables are less precise but increasingly useful:

- **Attention, focus, and workload monitoring**
  - Headsets can estimate **cognitive load, attention, and fatigue**, supporting applications in training, safety-critical work, and personalized learning. [8] [2] [3]
- **Neurofeedback and mental-health support**
  - Consumer and clinical EEG systems provide **real-time feedback** to help users regulate arousal, stress, or attention, used in clinics and some wellness contexts. [9] [3]
- **Gaming, VR, and creative tools**
  - BCIs enable **thought-based control in games and VR**, and experimental tools for **music, art, and design** driven by neural states. [9] [3]

What BCIs *cannot* reliably do yet

To keep expectations realistic:

- **Mind reading**: BCIs do not read arbitrary thoughts or memories; they decode specific, trained signals (e.g., “move cursor left,” “intend to type this letter”). [5] [1]
- **General-purpose telepathy**: There is no practical, high-bandwidth “thought-to-thought” communication between people; most communication still goes through text or speech decoders trained per user. [5]
- **Mass consumer augmentation**: Outside narrow medical and research use, BCIs are not yet mainstream for everyday productivity or enhancement; most consumer applications remain **monitoring, training, and basic interaction**, not full “control by thought.” [10] [2]

If you’d like, I can map these capabilities by BCI type (invasive vs non-invasive) and show which are realistically available in clinics vs still experimental.

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